Maria Diamanti

PhD

National Technical University of Athens, Iroon Polytechniou 9, Zografou, 15780

mdiamanti@netmode.ntua.gr | Google Scholar, Web of Science

Education

National Technical University of Athens (NTUA) *Athens*
School of Electrical and Computer Engineering
PhD *10/2018 - 10/2023*

- Major Areas: B5G/6G Systems, Wireless Networks, Resource Allocation, Optimization, Game Theory, Contract Theory, Reinforcement Learning
- Dissertation Title: "Resource Allocation and Incentive Mechanism Design in Next Generation Wireless Communication and Mobile Computing Networks"
- Advisor: Prof. Symeon Papavassiliou
- PhD Thesis Link

Aristotle University of Thessaloniki (AUTH) *Thessaloniki*
Department of Electrical and Computer Engineering
Diploma (M.Eng. equivalent) *09/2013 - 07/2018*

- Major Areas: Electronics and Computer Engineering
- Thesis Title: "Developing an Industrial Informatics Interface for Controlling a Robot Arm using a Microcomputer based on the IEC 61131-3 Standard"
- Advisor: Prof. Georgios Chasapis
- Grade: 8.07/10

Professional Experience

Research Committee (ELKE) of the National Technical University of Athens (NTUA) *Athens*

Principal Investigator

- PI for the project entitled "Taming System UnCErtainties via Efficient and ResiLiEnt Resource OrcheSTration in 3D NEtworks - CELESTE", HFRI (28615). *02/2026 - present*

Institute of Communication and Computer Systems (ICCS) of the National Technical University of Athens (NTUA) *Athens*

Postdoctoral Researcher

- Researcher for the project entitled "LEArning-Driven and Evolved Radio for 6G Communication Systems - 6G-LEADER", EU SNS-JU-2024 (101192080). *01/2025 - present*
- Researcher for the project entitled "HEXA-X-II: A Holistic Flagship towards the 6G Network Platform and System", EU SNS-JU-2022 (101095759). *01/2023 - 01/2025*

Research Committee (ELKE) of the National Technical University of Athens (NTUA) *Athens*

Researcher

- Researcher for the project entitled "Rethinking rEsource ALlocation in Interdependent wirelesS systeMs - REALISM", HFRI-FM17-2436. *12/2019 - 12/2022*
- Researcher for the project entitled "5G HEalth AquacultuRe and Transport Validation Trials - 5G-HEART", H2020- ICT-19-2019 (857034). *06/2019 - 11/2022*

Institute of Communication and Computer Systems (ICCS) of the National Technical University of Athens (NTUA) *Athens*

Researcher

- Researcher for the open call entitled "Computation Offloading for SMart tOuristic Sites - COSMOS", under "5GINFIRE" project, H2020-ICT-13-2016 (732497). *05/2019 - 07/2019*
- Researcher for the open call entitled "Autonomic CRoss layer protocol stack for SDR Systems - ACROSS", under "ORCA" project, H2020-ICT-13-2016 (732174). *10/2018 - 11/2018*

Awards & Scholarships

2026 **Best Paper Award**, 31st IEEE International Workshop on Computer-Aided Modeling and Design of Communication Links and Networks (CAMAD)

2025 **Rising Stars List for 2025**, Women in 6G

2023 **Ph.D. Thesis Award "Iakovos Gkiourounlian"**, Michael N. Stassinopoulos – VIOHALCO Public Benefit Foundation, Greece

2021 **HUAWEI Seeds for the Future Scholarship**, HUAWEI Greece

2020 **Scholarship for the support of the educational activities of NTUA during the academic year 2020**, Research Committee (ELKE) of the National Technical University of Athens (NTUA)

Teaching Experience

Spring 2026 **Stochastic Systems & Communications**, Undergraduate Course NTUA, Adjunct Lecturer

Spring 2021 **Stochastic Systems & Communications**, Undergraduate Course NTUA, Teaching Assistant

Spring 2021 **Queuing Systems**, Undergraduate Course NTUA, Teaching Assistant

Publications

Journal Publications

J30. P. Charatsaris, **M. Diamanti**, E. Stai, A. Zafeiropoulos, I. Dimolitsas, E. E. Tsiropoulou and S. Papavassiliou, "Anticipatory Computing in Space: Predictive Orchestration for Sustainable Satellite Systems," in IEEE Wireless Communications. *(Submitted)*

J29. F. Linardatou, **M. Diamanti**, E. E. Tsiropoulou and S. Papavassiliou, "On-Orbit Computing and Caching: A Distributed Regret Learning Approach," in IEEE Open Journal of the Communications Society, doi: 10.1109/OJCOMS.2026.3725112.

J28. P.Charatsaris, **M. Diamanti**, E. E. Tsiropoulou, and S. Papavassiliou, "On Efficient Uplink RSMA: A Genetic Approach for Power and Decoding Order Optimization," in IEEE Networking Letters. *(Under Review)*

J27. A. Sabyrbek, K. M. Rumman, D. Kaltsogiannis, **M. Diamanti**, P. Charatsaris, E. E. Tsiropoulou, and S. Papavassiliou, "AI-Native Integrated Sensing and Communication for Context-Aware Industrial IoT Coexistence," in IEEE Internet of Things Magazine. *(Accepted)*

J26. S. Barkatsa, P. Charatsaris, **M. Diamanti**, E. E. Tsiropoulou and S. Papavassiliou, "Proactive Cyber Deception in Attack Graphs: A Risk-aware Deployment Framework," in IEEE Transactions on Consumer Electronics, doi: 10.1109/TCE.2026.3714285.

J25. P. Charatsaris, **M. Diamanti**, A. Sabyrbek, E. E. Tsiropoulou and S. Papavassiliou, "Distributed Resource Control in Over-the-Air Computation based on Multi-Aggregative Games," in IEEE Transactions on Networking, doi: 10.1109/TON.2026.3748193.

J24. S. Voikos, P. Charatsaris, **M. Diamanti**, E. E. Tsiropoulou and S. Papavassiliou, "Radio and Compute Resource Allocation for SWIPT and RIS-assisted AirComp Federated Learning," in IEEE Transactions on Wireless Communications, doi: 10.1109/TWC.2026.3722443.

J23. P. Charatsaris, **M. Diamanti**, E. E. Tsiropoulou and S. Papavassiliou, "Balancing Accuracy-Latency Tradeoff in Hybrid Approximate-Delayed Computing Systems," in IEEE Transactions on Consumer Electronics, doi: 10.1109/TCE.2026.3681135.

J22. P. Charatsaris, S. Voikos, **M. Diamanti** and S. Papavassiliou, "Latency and Energy Minimization in RIS-enabled AirComp Federated Learning Networks," in IEEE Transactions on Vehicular Technology, doi: 10.1109/TVT.2026.3667419.

J21. P. Charatsaris, **M. Diamanti**, E. E. Tsiropoulou and S. Papavassiliou, "Risk-Conscious Computing in Space-Air-Ground IoT Networks: A Prospect-Theoretic Game Perspective," in IEEE Transactions on Network Science and Engineering, doi: 10.1109/TNSE.2025.3647157.

J20. I. Dimolitsas, **M. Diamanti**, S. Voikos and S. Papavassiliou, "Resilient RAN Selection and SFC Deployment in Dependable Wireless Edge Cloud Networks," in IEEE Transactions on Network and Service Management, doi: 10.1109/TNSM.2025.3645449.

J19. **M. Diamanti**, A. B. Rahman, P. Charatsaris, E. E. Tsiropoulou and S. Papavassiliou, "Resource Allocation and Pricing for Multi-Server Multi-Model Federated Learning based on Market Equilibrium," in Future Generation Computer Systems, vol. 175, pp. 108055, 2026, doi: 10.1016/j.future.2025.108055.

J18. I. Protogeros, **M. Diamanti**, D. Spatharakis and S. Papavassiliou, "Energy-Efficient and Data-Optimized Federated Learning for Distributed On-Device Intelligence," in IEEE Transactions on Consumer Electronics, doi: 10.1109/TCE.2025.3617645.

J17. O. U. Akgül, A. Varvara, A. De La Oliva, P. Charatsaris, **M. Diamanti**, P. Garau Burguera, M. Ericson, S. Wänstedt, M. Ziółkowski, H. Tarasiuk, H. Hellaoui, S. Papavassiliou, V. Tsekenis, S. Barbounakis, P.Demestichas, B. M. Khorsandi, and H. Karkous, "Sustainable 6G Architecture: An Organic Evolution of 5G Networks," in Computer Networks, pp. 111629, 2025, doi: 10.1016/j.comnet.2025.111629.

J16. G. Stavropoulou, E. Stai, **M. Diamanti**, and S. Papavassiliou, "Source-Rate Planning in Self-Powered Wireless Multi-Hop D2D Settings under Stochasticity: A Scenario-based Iterative Optimization Approach," in Computer Communications, pp. 108299, 2025, doi: 10.1016/j.comcom.2025.108299.

J15. P. Charatsaris, **M. Diamanti**, E. Stai, A. Zafeiropoulos and S. Papavassiliou, "From Ground to Orbit: A Reliable AI Framework for Unified Network Management," in IEEE Communications Standards Magazine, vol. 9, no. 2, pp. 48-55, 2025, doi: 10.1109/MCOMSTD.2025.3568900.

J14. M. S. Siraj, P. Charatsaris, **M. Diamanti**, E. E. Tsiropoulou, S. Papavassiliou, "HEROES: Humanitarian Emergency Response based on UAV-enabled Integrated Sensing and Communication, Positioning, and Satisfaction Games," in ACM Journal on Autonomous Transportation Systems, vol. 2, no. 4, pp. 1-26, 2025, doi: 10.1145/3715334.

J13. S. Barkatsa, **M. Diamanti**, P. Charatsaris, S. Voikos, E. E. Tsiropoulou and S. Papavassiliou, "Coordinated Jamming and Poisoning Attack Detection and Mitigation in Wireless Federated Learning Networks," in IEEE Open Journal of the Communications Society, vol. 6, pp. 3745-3759, 2025, doi: 10.1109/OJCOMS.2025.3558672.

J12. S. Barkatsa, **M. Diamanti**, P. Charatsaris, E. E. Tsiropoulou and S. Papavassiliou, "Jamming Attack Mitigation in Wireless Federated Learning Networks Using Bayesian Games," in IEEE Networking Letters, vol. 6, no. 4, pp. 247-251, Dec. 2024, doi: 10.1109/LNET.2024.3499360.

J11. **M. Diamanti**, N. Fryganiotis, S. Papavassiliou, C. Pelekis, E. E. Tsiropoulou, "Minimum Collisions Assignment in Interdependent Networked Systems via Defective Colorings," in ITU Journal on Future and Evolving Technologies, vol. 5, no. 1, pp. 1-12, 2024, 10.52953/JNDY5511.

J10. G. Kakkavas, **M. Diamanti**, V. Karyotis, K. N. Nyarko, M. Gabriel, A. Zafeiropoulos, S. Papavassiliou, K. Moessner, "5G Perspective of Connected Autonomous Vehicles: Current Landscape and Challenges Toward 6G," in IEEE Wireless Communications, vol. 31, no. 4, pp. 299-306, 2024, doi: 10.1109/MWC.014.2300277.

J9. P. Charatsaris, **M. Diamanti** and S. Papavassiliou, "Joint User Association and Resource Allocation for Hierarchical Federated Learning Based on Games in Satisfaction Form," in IEEE Open Journal of the Communications Society, vol. 5, pp. 457-471, 2024, doi: 10.1109/OJCOMS.2023.3347354.

J8. **M. Diamanti**, C. Pelekis, E. E. Tsiropoulou and S. Papavassiliou, "Delay Minimization for Rate-Splitting Multiple Access-Based Multi-Server MEC Offloading," in IEEE/ACM Transactions on Networking, vol. 32, no. 2, pp. 1035-1047, 2024, doi: 10.1109/ TNET.2023.3311131.

J7. **M. Diamanti**, G. Kapsalis, E. E. Tsiropoulou and S. Papavassiliou, "Energy-Efficient Rate-Splitting Multiple Access: A Deep Reinforcement Learning-Based Framework," in IEEE Open Journal of the Communications Society, vol. 4, pp. 2397-2409, 2023, doi: 10.1109/OJCOMS.2023.3322047.

J6. M. S. Siraj, A. B. Rahman, **M. Diamanti**, E. E. Tsiropoulou and S. Papavassiliou, "Alternative Positioning, Navigation, and Timing Enabled by Games in Satisfaction Form and Reconfigurable Intelligent Surfaces," in IEEE Systems Journal, vol. 17, no. 3, pp. 5035-5046, Sept. 2023, doi: 10.1109/JSYST.2023.3268989.

J5. S. Hossain, N. Irtija, **M. Diamanti**, F. Sangoleye, E. E. Tsiropoulou, S. Papavassiliou, "Reconfigurable Intelligent Surfaces-enabled Edge Computing: A Location-aware Task Offloading Framework," in ITU Journal on Future and Evolving Technologies, vol. 3, no. 3, pp. 830-843, 2022, doi: 10.52953/FLTJ9889.

J4. **M. Diamanti**, P. Charatsaris, E. E. Tsiropoulou and S. Papavassiliou, "Incentive Mechanism and Resource Allocation for Edge-Fog Networks Driven by Multi-Dimensional Contract and Game Theories," in IEEE Open Journal of the Communications Society, vol. 3, pp. 435-452, 2022, doi: 10.1109/OJCOMS.2022.3154536.

J3. G. Kakkavas, **M. Diamanti**, A. Stamou, V. Karyotis, F. Bouali, J. Pinola, O. Apilo, S. Papavassiliou, K. Moessner, "Design, Development, and Evaluation of 5G-Enabled Vehicular Services: The 5G-HEART Perspective," Sensors, vol. 22, no. 2, p. 426, Jan. 2022, doi: 10.3390/s22020426.

J2. **M. Diamanti**, P. Charatsaris, E. E. Tsiropoulou and S. Papavassiliou, "The Prospect of Reconfigurable Intelligent Surfaces in Integrated Access and Backhaul Networks," in IEEE Transactions on Green Communications and Networking, vol. 6, no. 2, pp. 859-872, June 2022, doi: 10.1109/TGCN.2021.3126784.

J1. **M. Diamanti**, G. Fragkos, E. E. Tsiropoulou and S. Papavassiliou, "Unified User Association and Contract-Theoretic Resource Orchestration in NOMA Heterogeneous Wireless Networks," in IEEE Open Journal of the Communications Society, vol. 1, pp. 1485-1502, 2020, doi: 10.1109/OJCOMS.2020.3024778.

Conference Publications

C35. F. Linardatou, **M. Diamanti**, P. Charatsaris, E. E. Tsiropoulou and S. Papavassiliou, "Sunlight-aware Task Offloading in LEO Satellite Networks: A No-Regret Learning Framework," ICC 2027 - IEEE International Conference on Communications, Washington, DC, USA. *(Submitted)*

C34. B. Ongdassynov, P. Charatsaris, **M. Diamanti**, E. E. Tsiropoulou and S. Papavassiliou, "Distributed Task Offloading for Low Altitude Economy Edge Computing," ICC 2027 - IEEE International Conference on Communications, Washington, DC, USA. *(Submitted)*

C33. S. Voikos, S. Barkatsa, P. Charatsaris, **M. Diamanti**, E. E. Tsiropoulou and S. Papavassiliou, "Power-Efficient Capacity Enhancement in RIS-assisted Semi-Grant-Free NOMA Networks," 2026 9th Conference on Cloud and Internet of Things (CIoT). *(Accepted)*

C32. P. Charatsaris, A. Sabyrbek, Y. Prathi, **M. Diamanti**, E. E. Tsiropoulou and S. Papavassiliou, "Multi-Agent Q-Learning for Power Control in AirComp-Based Traffic Monitoring Systems," 2026 IEEE 31th International Workshop on Computer Aided Modeling and Design of Communication Links and Networks (CAMAD). *(Accepted)*

C31. P. Charatsaris, **M. Diamanti**, A. Sabyrbek, E. E. Tsiropoulou and S. Papavassiliou, "Secure Wireless Communications in Concentrated Solar Power Fields Against Jamming Attacks," 2026 IEEE 31th International Workshop on Computer Aided Modeling and Design of Communication Links and Networks (CAMAD). *(Accepted)*

C30. **M. Diamanti**, P. Charatsaris, H. Al Kassir, M. Karavolos, N. Nomikos, T. Kobal, M. Andrews, D. Dinis, R. Wichman, S. Barkatsa and S. Papavassiliou, "Enhancing Spectral-Spatial-Temporal Efficiency of NOMA: Challenges and Design Considerations," 2026 22nd International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT), Reykjavik, Iceland, 2026, pp. 1352-1359, doi: 10.1109/DCOSS-IoT69657.2026.00201.

C29. M. S. Siraj, K. Azeez Syed, A. B. Rahman, **M. Diamanti** and E. E. Tsiropoulou, "A Hybrid Wi-Fi–ESB Communication Architecture for Scalable Distributed Counter-Unmanned Aircraft Systems," 2026 22nd International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT), Reykjavik, Iceland, 2026, pp. 545-552, doi: 10.1109/DCOSS-IoT69657.2026.00090.

C28. O. U. Akgül, V. Tsekenis, S. Barbounakis, P. Demestichas, P. Charatsaris, **M. Diamanti**, S. Papavassiliou, JJ van de Beek, A. Moradi, P. Gupta, P. Botsinis, A. Varvara, S. Eldessoki, M. Ziółkowski, H. Tarasiuk, K. Kuczyński, H. Harkous, P Garau Burguera and M. Ericson, "Sustainable 6G Architecture: A Hexa-X-II Perspective," 2026 22nd International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT), Reykjavik, Iceland, 2026, pp. 1304-1311, doi: 10.1109/DCOSS-IoT69657.2026.00194.

C27. S. Barkatsa, P. Charatsaris, **M. Diamanti** and S. Papavassiliou, "Secure and Power-Efficient Multi-User Scheduling in Semi-Grant-Free NOMA Networks: A Coalition Game Approach," International Conference on Game Theory and AI for Security, Springer Nature Switzerland, 2026, pp. 187-206, doi: 10.1007/978-3-032-08067-7_10.

C26. S. Barkatsa, **M. Diamanti**, P. Charatsaris and S. Papavassiliou, "Fair and Robust Federated Learning via Reputation-aware Incentives and Model Aggregation," 2025 IEEE 31st International Symposium on Local and Metropolitan Area Networks (LANMAN), Liile, France, 2025, pp. 1-6, doi: 10.1109/LANMAN66415.2025.11154591.

C25. P. Charatsaris, M. S. Siraj, **M. Diamanti**, E. E. Tsiropoulou and S. Papavassiliou, "Data Fusion and Resource Control for Multi-Receiver Over-the-Air Computation," IEEE Conference on Network Function Virtualization and Software Defined Networks (NFV-SDN) 2025, Athens, Greece, 2025. *(Accepted)*

C24. M. Saimler, S. Ickin, **M. Diamanti**, O. U. Akgül, G. Bernini, M. Zivkovic, and N. Toumi, "Advancing AI-Native 6G Networks by AI Enablers," IEEE International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC) 2025 Workshops, Istanbul, Türkiye, 2025. *(Accepted)*

C23. **M. Diamanti**, A. B. Rahman, P. Charatsaris, E. E. Tsiropoulou and S. Papavassiliou, "Resource Allocation as a Market: A Case Study on Multi-Server Multi-Model Federated Learning," 2025 20th Wireless On-Demand Network Systems and Services Conference (WONS), Hintertux, Austria, 2025, pp. 1-7.

C22. P. Charatsaris, **M. Diamanti**, E. E. Tsiropoulou and S. Papavassiliou, "Accuracy-Latency Tradeoff in Approximate and Delayed Computing as a Game in Satisfaction Form," ICC 2025 - IEEE International Conference on Communications, Montreal, QC, Canada, 2025, pp. 01-06, doi: 10.1109/ICC52391.2025.11161945.

C21. G. Mystriotis, M. S. Siraj, **M. Diamanti**, E. E. Tsiropoulou and S. Papavassiliou, "Autonomous Node Selection for Improved Accuracy in Multi-Model Federated Learning," 2024 IEEE Globecom Workshops (GC Wkshps), Cape Town, South Africa, 2024, pp. 1-6, doi: 10.1109/GCWkshp64532.2024.11101570.

C20. P. Charatsaris, A. S. Penafiel, **M. Diamanti**, E. E. Tsiropoulou and S. Papavassiliou, "Synergia: Device-Edge Server Association for ISAC-assisted Mobile Edge Computing Systems," GLOBECOM 2024 - 2024 IEEE Global Communications Conference, Cape Town, South Africa, 2024, pp. 61-66, doi: 10.1109/GLOBECOM52923.2024.10901178.

C19. G. Stavropoulou, E. Stai, **M. Diamanti** and S. Papavassiliou, "Network Operation Planning in Energy Harvesting Self-Powered Wireless Multi-hop Settings," 2024 20th International Conference on Wireless and Mobile Computing, Networking and Communications (WiMob), Paris, France, 2024, pp. 593-598, doi: 10.1109/WiMob61911.2024.10770455.

C18. P. Botsinis, M. Ericson, V. Tsekenis, M. Groshev, **M. Diamanti**, V. Sharma, S. Kuklinski, S. Eldessoki, A. Gündoğan, S. Barmpounakis, P. Demestichas, S. Papavassiliou, B. Mendes, J. Van De Beek, "New Access and Flexible Topologies in 6G: Architectural Implications," 2024 Joint European Conference on Networks and Communications & 6G Summit (EuCNC/6G Summit), Antwerp, Belgium, 2024, pp. 682-687, doi: 10.1109/EuCNC/6GSummit60053.2024.10597028.

C17. M. Saimler, S. Ickin, G. Bernini, N. Toumi, **M. Diamanti**, S. Papavassiliou, Ö. U. Akgül, B. M. Khorsandi, M. Zivkovic, "The Role of AI Enablers in Overcoming Impairments in 6G Networks," 2024 Joint European Conference on Networks and Communications & 6G Summit (EuCNC/6G Summit), Antwerp, Belgium, 2024, pp. 913-918, doi: 10.1109/EuCNC/6GSummit60053.2024.10597102.

C16. P. Charatsaris, M. Salcido, **M. Diamanti**, A. M. Ali, E. E. Tsiropoulou and S. Papavassiliou, "AGORA: A Multi-Provider Edge Computing Resource Management and Pricing Framework," 2024 International Wireless Communications and Mobile Computing (IWCMC), Ayia Napa, Cyprus, 2024, pp. 1685-1690, doi: 10.1109/IWCMC61514.2024.10592311.

C15. **M. Diamanti**, G. Kapsalis, E. E. Tsiropoulou and S. Papavassiliou, "Power and Rate Allocation for Energy-Efficient Rate-Splitting Multiple Access via Deep Q-Learning," 2023 IEEE Globecom Workshops (GC Wkshps), Kuala Lumpur, Malaysia, 2023, pp. 872-877, doi: 10.1109/GCWkshps58843.2023.10465023.

C14. P. Charatsaris, **M. Diamanti**, E. E. Tsiropoulou and S. Papavassiliou, "Efficient Power Control for Integrated Sensing and Communication Networks with Dual Connectivity," ICC 2023 - IEEE International Conference on Communications, Rome, Italy, 2023, pp. 5910-5915, doi: 10.1109/ICC45041.2023.10279737.

C13. P. Charatsaris, **M. Diamanti** and S. Papavassiliou, "On the Accuracy-Energy Tradeoff for Hierarchical Federated Learning via Satisfaction Equilibrium," 2023 19th International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT), Pafos, Cyprus, 2023, pp. 422-428, doi: 10.1109/DCOSS-IoT58021.2023.00073.

C12. G. Kakkavas, K. N. Nyarko, C. Lahoud, D. Kuehnert, P. Kueffner, M. Gabriel, S. Ehsanfar, **M. Diamanti**, V. Karyotis, K. Moessner, S. Papavassiliou, "Demo Proposal: Tele-Operated Support over 4G/5G Mobile Communications," 2021 IEEE International Mediterranean Conference on Communications and Networking (MeditCom), Athens, Greece, 2021, pp. 1-2, doi: 10.1109/MeditCom49071.2021.9647643.

C11. M. S. Siraj, A. B. Rahman, **M. Diamanti**, E. E. Tsiropoulou, S. Papavassiliou and J. Plusquellic, "Orchestration of Reconfigurable Intelligent Surfaces for Positioning, Navigation, and Timing," MILCOM 2022 - 2022 IEEE Military Communications Conference (MILCOM), Rockville, MD, USA, 2022, pp. 148-153, doi: 10.1109/MILCOM55135.2022.10017665.

C10. P. Charatsaris, **M. Diamanti**, E. E. Tsiropoulou and S. Papavassiliou, "Competitive Energy Allocation for Aerial Computation Offloading: A Colonel Blotto Game," GLOBECOM 2022 - 2022 IEEE Global Communications Conference, Rio de Janeiro, Brazil, 2022, pp. 970-975, doi: 10.1109/GLOBECOM48099.2022.10001655.

C9. **M. Diamanti**, N. Fryganiotis, S. Papavassiliou, C. Pelekis and E. E. Tsiropoulou, "On the Minimum Collisions Assignment Problem in Interdependent Networked Systems," 2022 IEEE Symposium on Computers and Communications (ISCC), 2022, pp. 1-6, doi: 10.1109/ISCC55528.2022.9912969.

C8. **M. Diamanti** and S. Papavassiliou, "Trading in Collaborative Mobile Edge Computing Networks: A Contract Theory-based Auction Model," 2022 IEEE 18th International Conference on Distributed Computing in Sensor Systems (DCOSS), 2022, pp. 387-393, doi: 10.1109/DCOSS54816.2022.00068.

C7. **M. Diamanti**, E. E. Tsiropoulou and S. Papavassiliou, "An Incentivization Mechanism for Green Computing Continuum of Delay-Tolerant Tasks," ICC 2022 - IEEE International Conference on Communications, 2022, pp. 3538-3543, doi: 10.1109/ICC45855.2022.9838752.

C6. G. Kakkavas, **M. Diamanti**, M. Gabriel, C. Lahoud, P. Akula, V. Karyotis, K. Moessner, S. Papavassiliou, "Demo Proposal: Tele-Operated Support over 4G/5G Mobile Communications," 2021 IEEE International Mediterranean Conference on Communications and Networking (MeditCom), 2021, pp. 1-2, doi: 10.1109/MeditCom49071.2021.9647643.

C5. G. Kakkavas, **M. Diamanti**, A. Stamou, V. Karyotis, S. Papavassiliou, F. Bouali, K. Moessner, "5G Network Requirement Analysis and Slice Dimensioning for Sustainable Vehicular Services," 2021 17th International Conference on Distributed Computing in Sensor Systems (DCOSS), 2021, pp. 495-502, doi: 10.1109/DCOSS52077.2021.00082.

C4. **M. Diamanti**, E. E. Tsiropoulou and S. Papavassiliou, "The Joint Power of NOMA and Reconfigurable Intelligent Surfaces in SWIPT Networks," 2021 IEEE 22nd International Workshop on Signal Processing Advances in Wireless Communications (SPAWC), 2021, pp. 621-625, doi: 10.1109/SPAWC51858.2021.9593111.

C3. **M. Diamanti**, M. Tsampazi, E. E. Tsiropoulou and S. Papavassiliou, "Energy Efficient Multi-User Communications Aided by Reconfigurable Intelligent Surfaces and UAVs," 2021 IEEE International Conference on Smart Computing (SMARTCOMP), 2021, pp. 371-376, doi: 10.1109/SMARTCOMP52413.2021.00075.

C2. **M. Diamanti**, E. E. Tsiropoulou and S. Papavassiliou, "Resource Orchestration in UAV-assisted NOMA Wireless Networks: A Labor Economics Perspective," ICC 2021 - IEEE International Conference on Communications, 2021, pp. 1-6, doi: 10.1109/ICC42927.2021.9500715.

C1. **M. Diamanti**, G. Fragkos, E. E. Tsiropoulou, S. Papavassiliou, "Resource Orchestration in Interference-Limited Small Cell Networks: A Contract-Theoretic Approach," International Conference on Network Games, Control and Optimization (NETGCOOP), Springer, Cham, 2021, pp. 101-109, doi: 10.1007/978-3-030-87473-5_10.

Book Chapters

B2. **M. Diamanti**, P. Charatsaris, E. Stai, A. Zafeiropoulos, S. Papavassiliou, "Reliable AI for Decision-Making and Resource Allocation in Integrated Terrestrial and Non-Terrestrial Networks," in Artificial Intelligence for Integrated Terrestrial and Non-Terrestrial Networks, Springer, Cham, 2026, doi: 10.1007/978-3-032-06512-4_4.

B1. M. Ericson et al., "Architecture Enablers for 6G," in 6G Systems and Technologies, edited by M. A. Uusitalo, P. Rugeland, M. R. Boldi, and A. Nimr, Wiley, 2026, pp. Ch. 5. doi: 10.1002/9781394363605.ch5.

Presentations

Tutorials

M. Diamanti, S. Papavassiliou, "Resource Orchestration in 3D Networks under Stochastic Dynamics," 2026 29th Conference on Innovation in Clouds, Internet and Networks (ICIN), Athens, Greece, 2026.

M. Diamanti, C. Pelekis, S. Papavassiliou, "Distributed Resource Orchestration in Realistic 6G Wireless Systems: From Practice, to Theory, to Practice," IEEE Symposium on Computers and Communications (ISCC), Rhodes, Greece, 2022.

M. Diamanti, C. Pelekis, S. Papavassiliou, "Economic-driven Resource Management in Next Generation (NextG) Communications and Computing Systems," 2022 IEEE International Mediterranean Conference on Communications and Networking (MeditCom), Athens, Greece, 2022.

M. Diamanti, C. Pelekis, S. Papavassiliou, "Resource Management under Uncertainties, Risks and Information Incompleteness in Next Generation (NextG) Networks," 2022 18th International Conference on Network and Service Management (CNSM), Thessaloniki, Greece, 2022.

Communications & Standards Summits

M. Saimler, G. Bernini, N. Toumi, S. Ickin, **M. Diamanti**, M. Zivkovic "Transforming ICT with Cutting-Edge AI Enablers: A 6G Perspective," ETSI Artificial Intelligence (AI) Conference - Status, Implementation and Way Forward of AI Standardization, Sophia Antipolis, France, 2024.

M. Saimler, S. Ickin, H. Larsson, M. Ericson, B. Masood Khorsandi, O. Akgul, G. Bernini, M. Zivkovic, S. Kuklinski, **M. Diamanti**, M. A. Habibi, B. Mendes "Realistic Usage of AI in 6G Networks: Overcoming Impairments with AI," Towards Realistic Usage of AI in 6G Networks Workshop, Lisbon, Portugal, 2023.

Student Supervision

PhD Students

- Filothei Linardatou *(2026-present)*
  Topic: Optimization of Resilient Networked Systems under Uncertainties
- Stefanos Voikos *(2025-present)*
  Topic: Sixth Generation Terrestrial/Non-Terrestrial Network Integration
- Sofia Barkatsa *(2025-present)*
  Topic: Resilience and Reliability in AI-enabled Secure Dynamic Mobile Computing Systems
- Panagiotis Charatsaris *(2021-present)*
  Topic: Compute and Radio Resource Orchestration and Data Fusion in NGMA 6G-enabled Computing Paradigms

Master Students

- Dimitrios Thanogiannis *(2026)*
  Topic: Honeypot Deployment in Attack Graphs via Hypergames with Asymmetric Information

- Konstantinos Tziapouras *(2026)*
  Topic: Game-Theoretic Distributed Placement of Service Function Chains in Edge Computing Networks
- Georgios Stathatos *(2026)*
  Topic: Optimization and Experimental Evaluation of Non-Orthogonal Multiple Access in a 5G Infrastructure
- Stella Kostaki *(2026)*
  Topic: Dynamic Hierarchical Cell-Switching in Integrated Terrestrial and Non-Terrestrial Networks
- Thanasis Konstantopoulos *(2026)*
  Topic: Realistic Deployment and Theoretical Validation of Honeypot Systems using BloodHound
- Georgios Stroumpakos *(2025)*
  Topic: Computation Task Offloading and Resource Allocation in Semi-Grent-Free NOMA Networks
- Eleutheria Kalogianni *(2025)*
  Topic: Security Function Chain Placement in Edge-Cloud Attack Graphs using Game Theory
- Filothei Linardatou *(2025)*
  Topic: Task Offloading and Caching in Satellite Networks via Stochastic Games with Incomplete Information
- Dimitrios Koumoustiotis *(2025)*
  Topic: Joint Task Offloading and Power Control in Dynamic Low-Earth-Orbit Satellite Networks: A Hierarchical Game
- Nikitas Chatzis *(2025)*
  Topic: Service Function Chain Placement and Resource Allocation in Terrestrial/Non-Terrestrial Networks
- Stefanos Voikos *(2025)*
  Topic: Resource Allocation in Wireless Federated Learning Networks with Wireless Energy Transfer
- Christos Bovolis *(2025)*
  Topic: Load Balancing in Satellite Networks Using Game Theory and Regret Learning
- Georgios Mistriotis *(2025)*
  Topic: Design of Matching Algorithms for Optimizing the Performance of Multi-Model Federated Learning
- Sofia Barkatsa *(2024)*
  Topic: Bayesian Games in Wireless Federated Learning Networks with Malicious Users
- Ioannis Protogeros *(2024)*
  Topic: Resource Allocation and Importance-Based Data Selection in NOMA Wireless Federated Learning Networks
- Michalis Vassilakos *(2024)*
  Topic: Deep Reinforcement Learning-based Optimization for Secure Data Offloading in Wireless Networks
- Vasileia Ntaliani *(2024)*
  Topic: Hierarchical Games in Wireless Integrated Access and Backhaul Networks for Federated Learning
- Symeon Vaidanis *(2023)*
  Topic: Optimization of Reconfigurable Intelligent Surfaces for Integrated Sensing and Communication Networks
- Andreas Kalogeras *(2023)*
  Topic: Game Theory Applications in Reconfigurable Intelligent Surface-enabled Communication Systems
- Georgios Kapsalis *(2022)*
  Topic: Resource Allocation in Rate-Splitting Multiple Access Networks Using Reinforcement Learning
- Maria Tsampazi *(2021)*
  Topic: Energy-Efficient Multi-User Communications Aided by Reconfigurable Intelligent Surfaces and UAVs
- Panagiotis Charatsaris *(2021)*
  Topic: 6G Integrated Access and Backhaul Networks with Reconfigurable Intelligent Surfaces

Outreach & Professional Development

Workshop Program Co-Chair

2026 • **6th International Workshop on Real-life Modeling in 6G Networks (REFRESH)**, Reykjavik, Iceland, 2026.

Conference Publicity Co-Chair

2027
- **33nd Annual International Conference On Mobile Computing And Networking (MobiCom)**, 2027.
- **32th IEEE International Workshop on Computer-Aided Modeling and Design of Communication Links and Networks (CAMAD)**, London, United Kingdom, 2027.

2026
- **17th IFIP Wireless and Mobile Networking Conference (WMNC)**, Enschede, Netherlands, 2026.
- **27th IEEE International Conference on High Performance Switching and Routing (HPSR)**, Montreal, Canada, 2026.
- **9th Conference on Cloud and Internet of Things (CIoT)**, Athens, Greece, 2026.

2025
- **16th Conference on Game Theory and AI for Security (GameSec)**, Athens, Greece, 2025.
- **IEEE International Workshop on Computer-Aided Modeling and Design of Communication Links and Networks (IEEE CAMAD)**, Phoenix, Arizona, USA, 2025.
- **16th IFIP Wireless and Mobile Networking Conference (WMNC)**, Leuven, Belgium, 2025.
- **IEEE International Symposium on Local and Metropolitan Area Networks (LANMAN)**, Lille, France, 2025.
- **8th International Balkan Conference on Communications and Networking (BalkanCom)**, Athens, Greece, 2025.

2024
- **7th Conference on Cloud and Internet of Things (CIoT)**, Montreal, Canada, 2024.
- **15th IFIP Wireless and Mobile Networking Conference (WMNC)**, Venice, Italy, 2024.
- **21st Annual IEEE International Conference on Sensing, Communication, and Networking (SECON)**, Phoenix, USA, 2024.

Conference Session Chair

2026 **9th Conference on Cloud and Internet of Things (CIoT)**, Athens, Greece, 2026.

2024 **29th IEEE Symposium on Computers and Communications (ISCC)**, Paris, France, 2024.

2022 **31st International Conference on Computing and Communication Networks (ICCCN)**, Virtual, 2022.

Conference Program Committee Member

- IEEE ICC 2024, 2027
- IEEE SECON 2024, 2026
- IEEE IFIP Networking 2024
- IEEE GLOBECOM 2022, 2023, 2024, 2025, 2026
- IEEE ISCC 2021, 2022, 2023, 2024, 2025, 2026
- IEEE WoWMoM 2022, 2023, 2024, 2025, 2026, 2027
- IEEE CIoT 2024, 2025, 2026
- WiMob 2022, 2023, 2024, 2025, 2026
- WiOpt 2025, 2026, 2027
- IEEE WMNC 2021, 2022, 2024, 2025, 2026
- IEEE WCNC 2025
- IEEE LANMAN 2025, 2026
- IEEE CAMAD 2025

- IEEE WiSEE 2026
- IEEE DCOSS-IoT 2027
- ACM ICDCN 2024, 2025
- ACM MobiArch 2026
- GameSec 2025, 2026
- COMSNETS Workshops 2024, 2025, 2026
- IEEE HPSR Workshops 2023
- IEEE MeditCom Workshops 2022
- ICDCN 2024, 2025, 2026, 2027
- ICCCN 2025, 2026
- EAI SmartGIFT 2026

Journal Reviewer

- IEEE Transactions on Communications
- IEEE Transactions on Mobile Computing
- IEEE Transactions on Vehicular Technology
- IEEE Open Journal of the Communications Society
- IEEE Journal on Selected Areas in Communications
- IEEE Transactions on Machine Learning in Communications and Networking
- IEEE Transactions on Cognitive Communications and Networking
- IEEE Transactions on Green Communications and Networking
- IEEE Transactions on Network and Service Management
- IEEE Transactions on Consumer Electronics
- IEEE Transactions on Circuits and Systems
- IEEE/ACM Transactions on Networking
- IEEE Systems Journal
- IEEE Communications Letters
- IEEE Networking Letters
- IEEE Vehicular Technology Magazine
- IEEE Communications Magazine
- IEEE Wireless Communications
- IEEE IT Professional
- Computer Networks
- Computer Communications
- Pervasive and Mobile Computing
- Physical Communication
- Future Generation Computer Systems
- Journal of Cloud Computing
- Journal of Network and Systems Management

Professional Memberships

IEEE Member

Technical Chamber of Greece

Language Skills

English: C2 (Full Professional Proficiency)

German: B2 (Elementary Proficiency)

Greek: Native